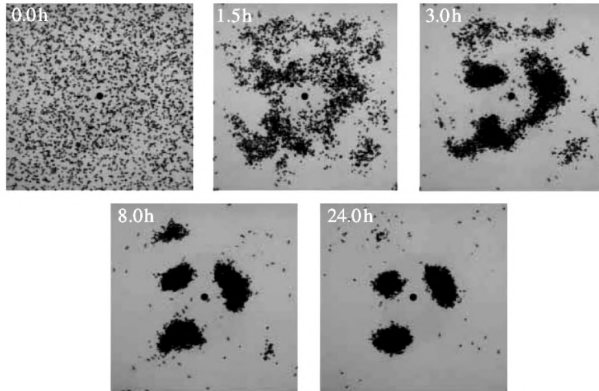


Modelling and Simulation

Lecture 8: Agent Based Models

1. Motivation

Ant Corpse Piles (*Messor sanctus*)

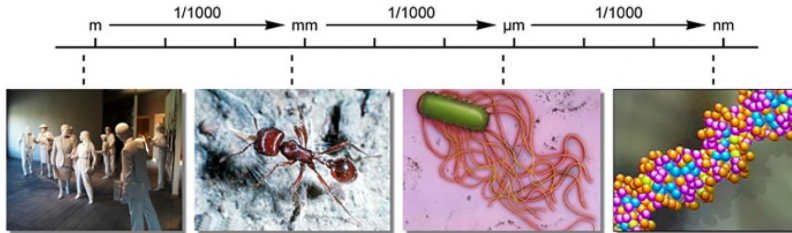


Jost *et al.*, J.R.Soc. Interface, 2007

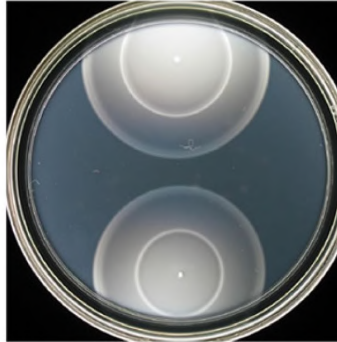
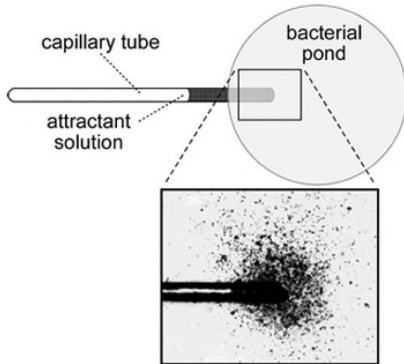
Ant corpse piles: Some questions

- ▶ How does it work ?
- ▶ Is it swarm intelligence ?
- ▶ What is the simplest model able to explain the process ?

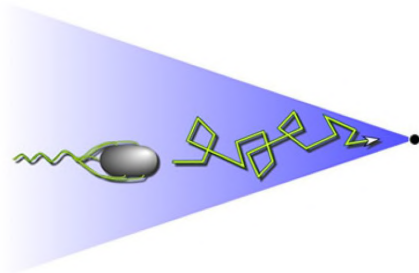
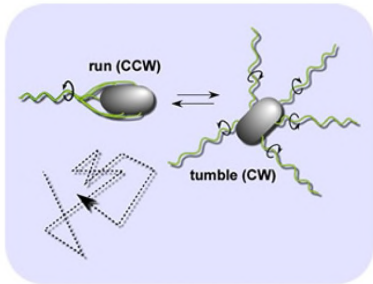
Bacteria



Chemotaxy



Movement



Stock Exchange

- ▶ Each trader makes individual decisions about bids and asks
- ▶ Most traders follow individual strategies
- ▶ Is it possible to explain market evolution knowing individual strategies.

Paul Jorion, 2007, *Adam Smiths Invisible Hand Revisited. An Agent-Based simulation of the New York Stock Exchange*,

http://www.pauljorion.com/blog/wp-content/uploads/2007/04/adamsmith-kyoto_rev.pdf

Agent based models

Main idea: Modeling the basic entities as individuals and observe the global *emergent* behavior.

Many more examples:

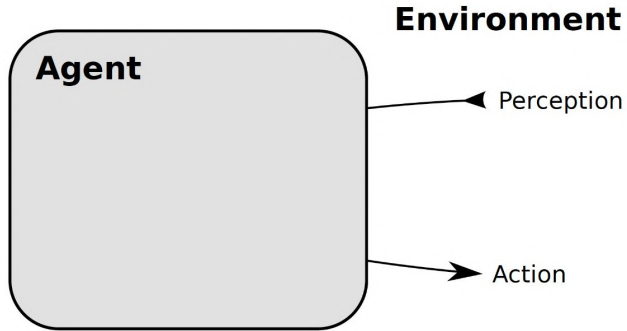
- ▶ Pedestrian simulation
- ▶ Epidemy propagation
- ▶ Ecological modeling
- ▶ ...

2. Agents

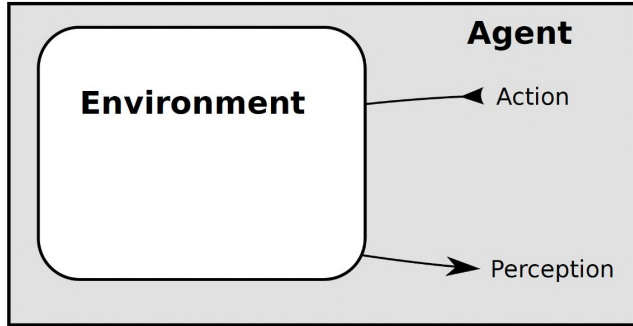
(Intelligent) Agents

- ▶ Agents are the fundamental entities of ABM
- ▶ Concept introduced in the Artificial Intelligence field
- ▶ Autonomous and decentralized
- ▶ Interact with an environment

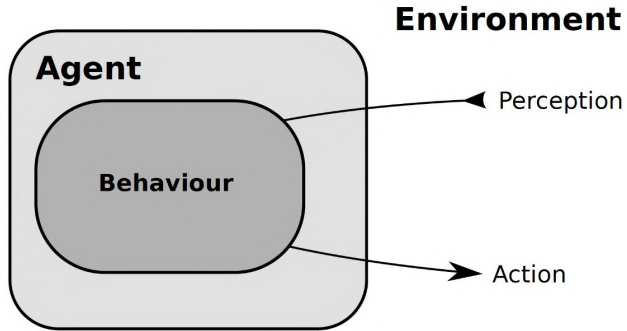
Agent



From an agent point of view



Simple Reflex Agent



Simple Reflex Agent

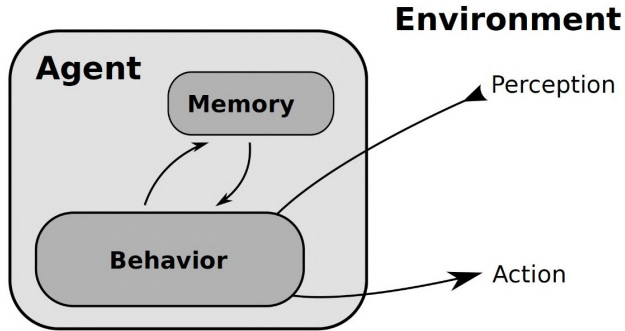
Behaviour: how the agent will react to the environment *perception*.

- ▶ Usually rule-based

PERCEPTION $\rightarrow$ ACTION

- ▶ May be stochastic
- ▶ Perception/Knowledge of environment is limited
- ▶ The *action* may affect the environment

Intelligent Agent



Intelligent Agent

The agent has a **state**, which can be as simple as a boolean or as complex as it needs to be

Behaviour function:

$$\text{PERCEPTION} \times \text{STATE} \rightarrow \text{ACTION} \times \text{STATE}$$

- ▶ The state is a kind of memory of past perceptions/actions
- ▶ The behaviour depends on memory
- ▶ Hence the agent is capable of **learning**

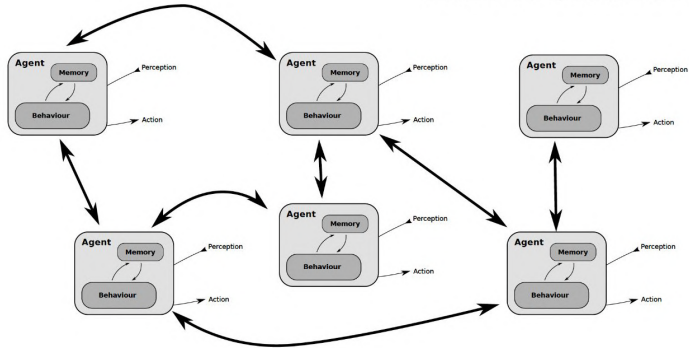
Example: Trading agent

```
def behavior( price, state ):  
  
    lastTxPrice, cash, stocks = state  
  
    if (price > lastTxPrice) and (price - lastTxPrice ) > RL:  
n = floor( stocks * Cs )  
return SELL( n, price )  
    elif (price < lastTxPrice) and (lastTxPrice - price) > RL:  
n = floor( ( cash * Cb ) / price )  
return BUY( n, price )  
    else:  
return NOP
```

3. Multi-Agent systems

Multi-Agent Systems

Environment



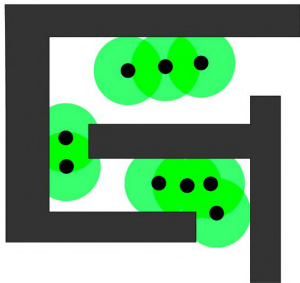
Multi-Agent Systems

- ▶ Usually a single system is modeled by many agents
- ▶ They could be identical, or similar, or not. . .
- ▶ They interact, either through the environment or directly.
- ▶ Multi-agent systems are not synonymous of ABM. For instance:
 - ▶ Optimization
 - ▶ Network security
 - ▶ Videogames

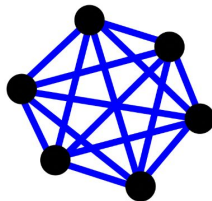
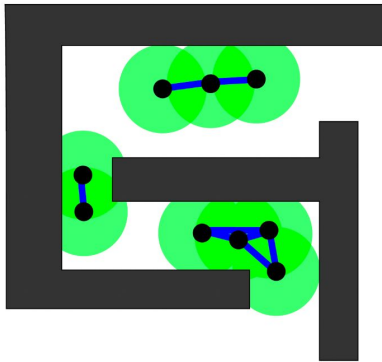
Spatialized Agents (Physical)

- ▶ Agent may have a spatial location (2D, 3D, graph)
- ▶ They may move across the domain as a result of their actions (mobile)
- ▶ The location of the agents may affect:
 - ▶ Their environment perception
 - ▶ Their interactions with other agents

Environment awareness



Interaction topology



Agent Based Models

- ▶ The use of a multi-agent system system to model a natural phenomenon.
- ▶ A complex collective behaviour **emerges** from the simple behaviours of agents.

4. Implementation of Agent Based Models

Object-Oriented definition

- ▶ **Class:** type of agent
- ▶ **Instance:** Agent
- ▶ **Private members:** internal state
- ▶ **Public methods:** behaviour

```
class FooAgent {  
    private final long ID = 223;  
    private int count = 2;  
    private double ratio = 1.5;  
    public Action behaviour( Perception p ) {  
        //...  
    }  
}
```

Asynchronous update

```
agents, env = initialize()
t = t_init
while t < t_max:
    for agent in agents:
        p = computePerceptionFor(agent, env, agents)
        action = agent.behaviour(p)
        updateEnvironment( env, action )
        increment(t)
```

Synchronous update

```
agents, env = initialize()
t = t_init
while t < t_max:
    ps = computeAllPerceptions(env, agents)
    actions = allBehaviours( agents, ps )
    updateEnvironment( env, actions )
    increment(t)
```

Lagrangian approach

- ▶ Common approach
- ▶ Each agent is aware of its location
- ▶ Interactions and environment awareness can be globally computed.

```
agents = [  
    Agent( id=1, posX=8.2, posY=0.5, ... ),  
    Agent( id=2, posX=9.1, posY=2.7, ... ),  
    Agent( id=3, posX=4.6, posY=1.8, ... ),  
    ...  
]
```

Spatial optimisation

- ▶ In most lagrangian models where agents communicate locally, it may be expensive to compute the interaction network.
- ▶ Naive approach, $O(n^2)$.
- ▶ Some specialised data structure may speed-up the process.
- ▶ For instance **k-d trees**:
 - ▶ Construction: $O(n \log n)$
 - ▶ Range search (in 2D): $O(n\sqrt{n})$

Eulerian approach

- ▶ Environment is a regular grid of cells
- ▶ Each cell contains a list of agents

Advantages:

- ▶ Interaction network easy to compute (neighboring cells).
- ▶ Interactions are local (parallelism possible)

Disadvantages:

- ▶ Loss of spatial precision

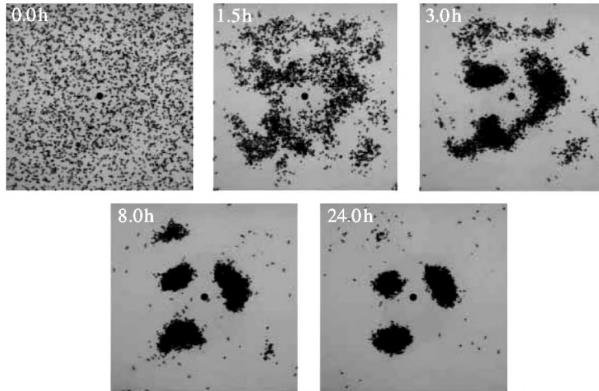
Time

- ▶ Usually continuous time
- ▶ But ABM can be used inside a Discrete Event System to update its state and produce new events.

```
def behavior( event, state ) :  
    ...  
    return newState, [events]
```

5. Ant corpse clustering

Ant Corpse Piles (*Messor sanctus*)



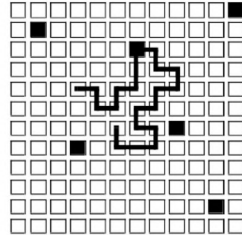
Jost *et al.*, J.R.Soc. Interface, 2007

Questions

- ▶ How does it work ?
- ▶ Is it swarm intelligence ?
- ▶ What is the simplest model able to explain the process ?

Deneubourg's Model (1991)

- ▶ Ants on a regular grid, with 4 directions
- ▶ Random walk, can walk over a corpse
- ▶ Sequential (asynchronous) updating scheme



Ant Behaviour (i)

- ▶ With propability P_p , the workers pick up a corpse if it is isolated or in a small cluster
- ▶ With probability P_d , the workers deposit a corpse in large cluster of dead bodies
- ▶ How the ant does evaluate the cluster size ?
 - ▶ Each ant has a memory M of size n :
 - ▶ The memory locations indicate the state of the cells visited by the ant during the last n steps: $M(i) = 1$ if there was a corps at time $t - i$, 0 otherwise

Ant Behaviour (ii)

- The probabilities are computed at each step as:

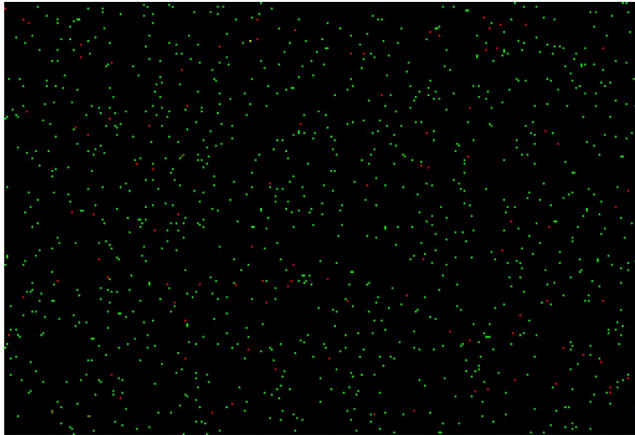
$$f = \sum_{i=1}^n M(i)$$

$$P_p = \left(\frac{k_1}{k_1 + f} \right)^2$$

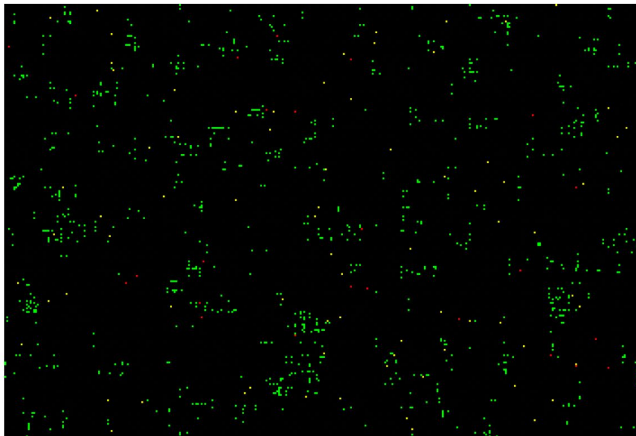
$$P_d = \left(\frac{f}{f + k_2} \right)^2$$

where k_1 and k_2 are model parameters.

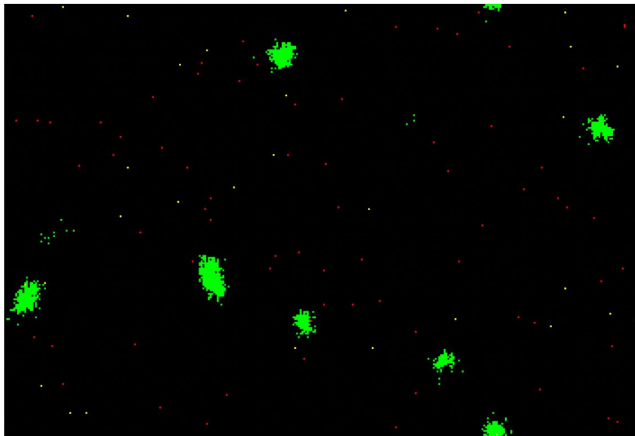
Result (i)



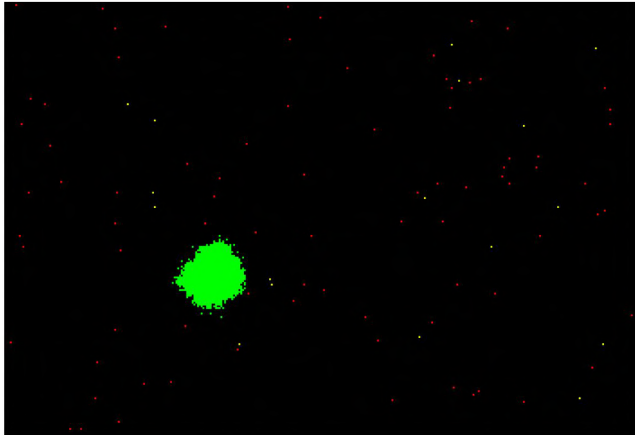
Result (ii)



Result (iii)



Result (iv)

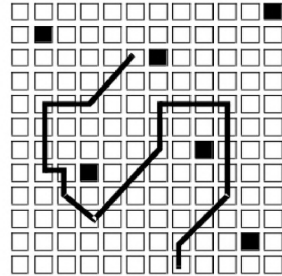


Smart ants

- ▶ Deneubourg's model works well
- ▶ Basic mechanism is intuitive
- ▶ But it requires a lot of "intelligence" from ants
- ▶ What about dumber ants ?

Unige Model

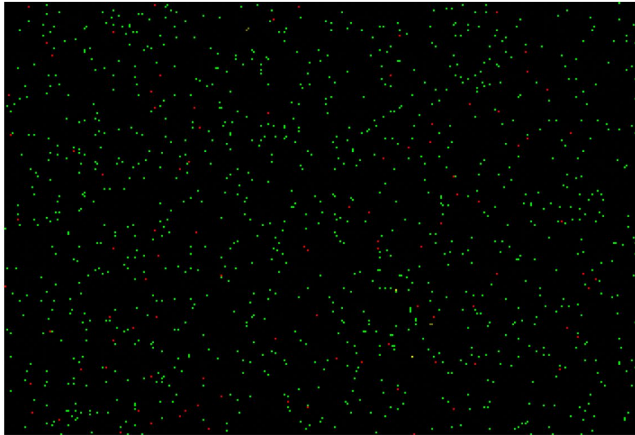
- ▶ Regular Grid, with 8 directions
- ▶ Random Walk with large diffusion constant
- ▶ Asynchronous updating



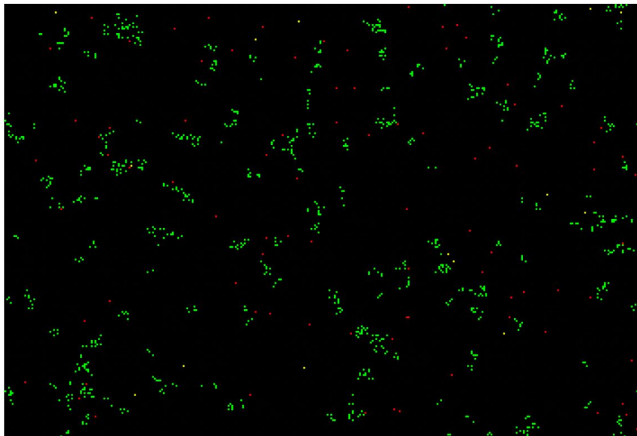
Behavior

- ▶ The ants avoids all obstacles:
 - ▶ ant corpses
 - ▶ other working ants
 - ▶ boundaries and walls
- ▶ An unloaded ant always picks a found corpse
- ▶ A loaded ant who finds another corpse always drops the carried corpse.

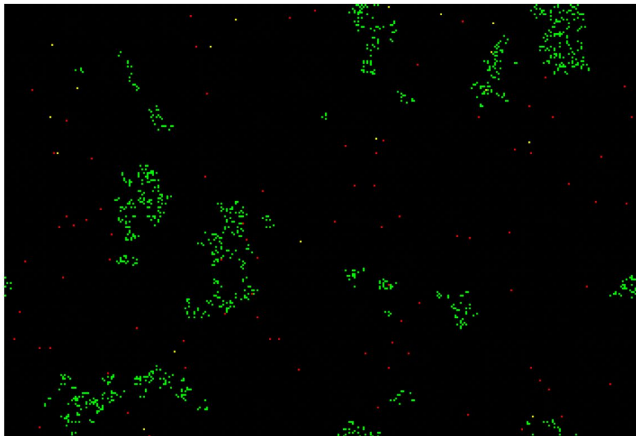
Result (i)



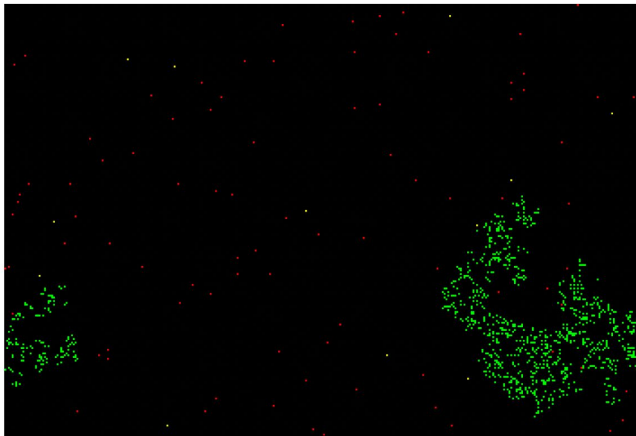
Result (ii)



Result (iii)



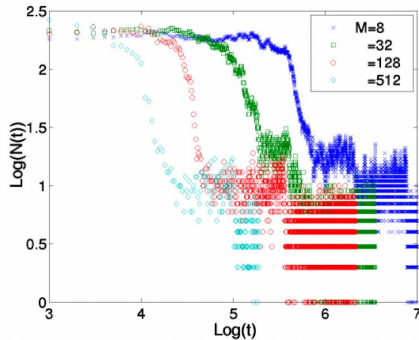
Result (iv)



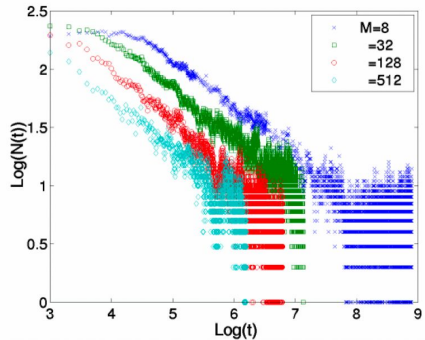
It works... but why ?

- ▶ The probabilities to remove a corpse from a cluster, or to add a new corpse are the same.
- ▶ Ants make no difference between a large or a small cluster
- ▶ When a cluster is emptied it will never reappear.
- ▶ Due to fluctuations, all clusters but one will eventually reach a zero size

Quantitative results



Deneubourg, with 8 directions

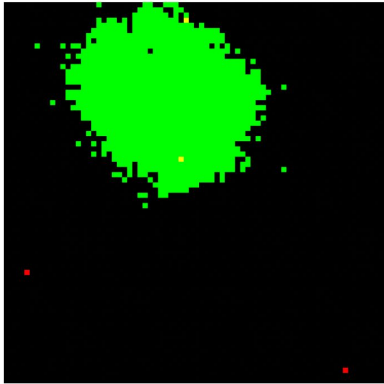


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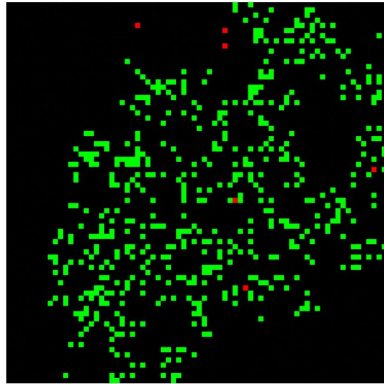
Quantitative results

- ▶ In Deneubourg's model converges $\sim 10x$ faster (using better random walk).
- ▶ In both models: not a collective behavior, $N(t) = f(Mt)$
- ▶ One single ant would make it, but slower

Final Cluster

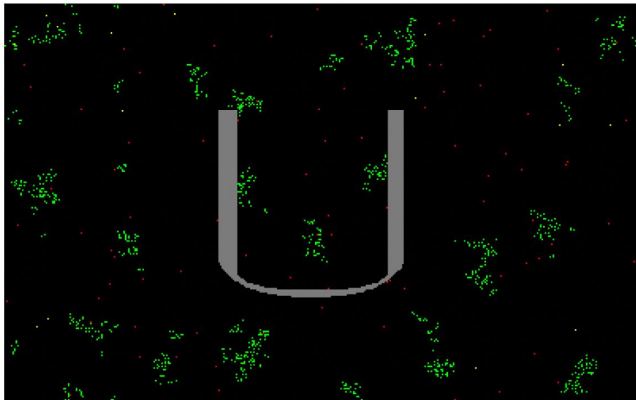


Deneubourg, with 8 directions

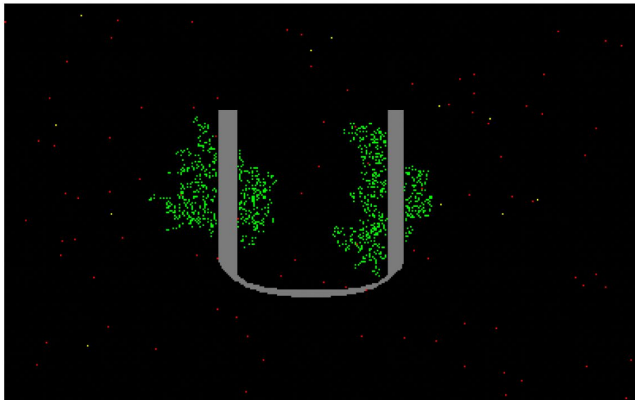


Unige

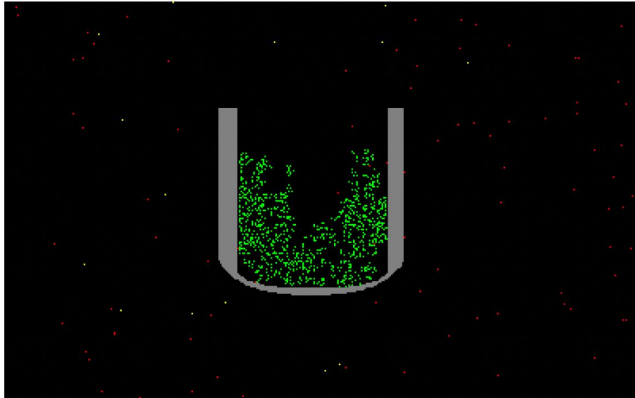
With obstacle (i)



With obstacle (ii)



With obstacle (iii)

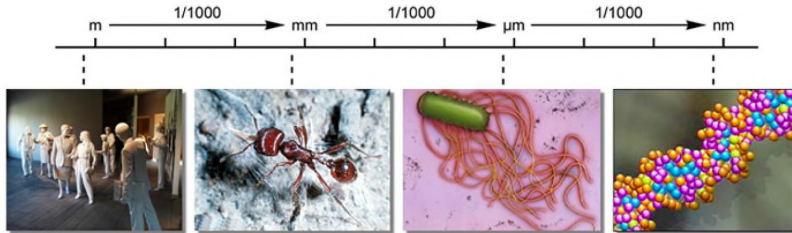


Conclusions

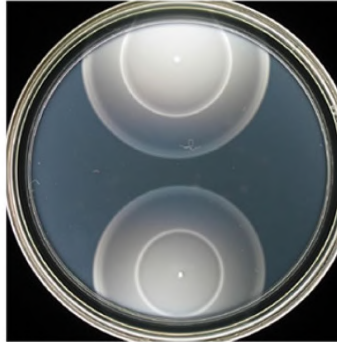
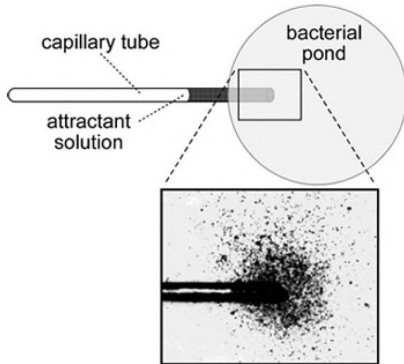
- ▶ Ant corps pile construction can be explained by statistical fluctuations
- ▶ Yet, intelligence speeds up the process
- ▶ Not a collective effect, just a collaboration with a linear speedup

6. Bacteria chemotaxy

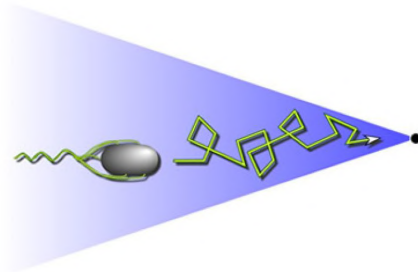
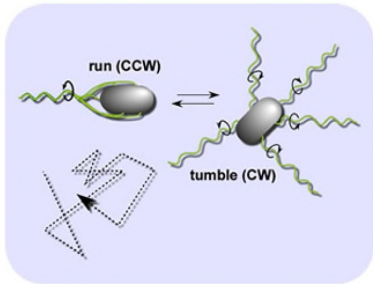
Bacteria



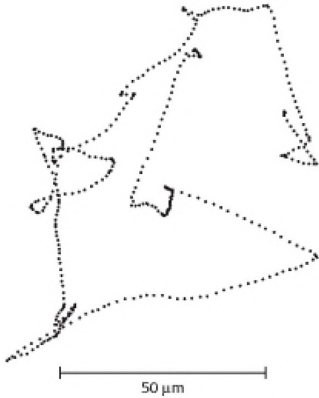
Chemotaxy



Movement (1)



Movement (2)



Model

- ▶ Eulerian 2D Grid
- ▶ In each cell (x, y) we have:
 - ▶ List of Bacteria in the (x, y)
 - ▶ Concentration of nutrient $\rho_{x,y}$
- ▶ Bacteria are agents i with state (d_i, m_i) :
 - ▶ d_i : last direction taken (N, S, E, W)
 - ▶ m_i : last concentration of nutrient

Behavior

- ▶ Bacteria remember last concentration (d_i)
- ▶ Bacteria at position (x, y) perceive the current concentration $\rho_{x,y}$
- ▶ There are two model parameters:
 - ▶ p_i : probability of tumbling when concentration increases
 - ▶ p_d : probability of tumbling when concentraion decreases
 - ▶ with $p_d > p_i$

Behavior function

```
def behaviour( rho, m_i, d_i ):  
    if rho <= m_i:  
        p = p_d  
    else:  
        p = p_i  
    if random() <= p:  
        return rho, randomDirection()  
    else:  
        return rho, d_i
```

Model



Environment

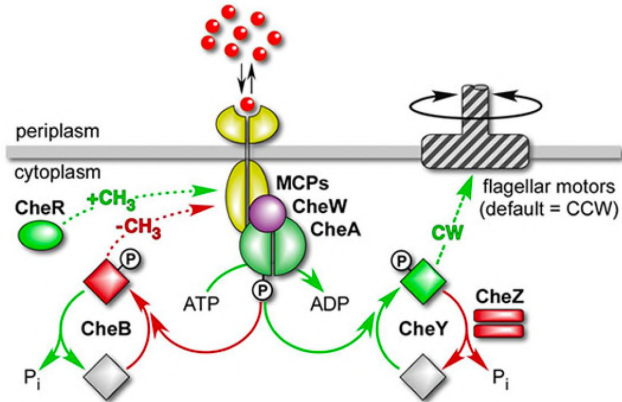
Nutrient diffusion: (solved with finite differences)

$$\frac{\partial \rho_{x,y}(t)}{\partial t} = D \nabla^2 \rho_{x,y}(t)$$

Bacteria movement:

- Each bacteria i is moved to the next cell in the direction d_i

Molecular Mechanism



See also

- ▶ An overview of E. coli chemotaxis
- ▶ Robustness in bacterial chemotaxis, Alon *et al.*, Nature **397**, 1999

<https://www.youtube.com/watch?v=Hc6knq5A8lQ>